NYC 311 Complaints Analysis – Manhattan (2025)

NYC 311 neighborhood issues

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Project Links:
Website: https://groupproject311.streamlit.app/
Github: https://github.com/QMSS-G5063-2026/Group_Project_311_Service_Requests/tree/main
Data Source: NYC Open Data sources - 311 Service Requests Query

Project Overview:
This project analyzes NYC 311 service request data, focusing on Manhattan in 2025. The goal is to explore patterns in public complaints across categories, locations, and time to better understand quality-of-life issues in the city.

We built an interactive Streamlit dashboard that allows users to:

- Explore complaint categories and subcategories
- Analyze trends over time
- Visualize geographic patterns
- Identify dominant issues across neighborhoods

Data and Preprocessing:

- From the original dataset, which included records from 2024 and 2025 across all New York City boroughs, we narrowed the scope to only 2025 data for Manhattan to make the analysis more computationally manageable and focused. This filtering allowed us to work with a more targeted subset while still capturing meaningful patterns in 311 service requests.
- To enhance data quality, we enriched the dataset using geospatial processing. The 311 records were converted into a spatial format using latitude and longitude coordinates, and GeoJSON boundary files were used to map each record to its corresponding neighborhood. Missing or invalid ZIP codes were then backfilled through spatial joins with ZIP code boundary data, ensuring more complete and accurate location information for each complaint.
- Finally, we created a custom complaint grouping variable to simplify analysis. Detailed complaint types were mapped into broader categories such as Noise Issues, Sanitation & Waste, Public Safety & Illegal Activity, and others. This categorization enabled higher-level trend analysis while still preserving the granularity of the original complaint data for more detailed exploration if needed.

Dashboard Components:

- Complaint Category Explorer (Home Page)
- Displays main complaint groups as interactive cards

- Users can click categories to drill down into:
- Complaint types
- Detailed complaint breakdowns
- Provides an overview of the dominant issues in Manhattan

Map Visualization

- Displays 311 complaint locations using latitude and longitude coordinates across Manhattan
- Highlights spatial clustering of issues through a heatmap (for overall trends) and point markers (for filtered views)
- Includes interactive filtering by neighborhood, ZIP code, complaint category, and specific issue, consistent with other pages
- Provides key summary metrics (hotspot neighborhood, total reports, top issue) to contextualize the map insights

Over Time Analysis

- Monthly aggregation of complaints using the Month variable
- Interactive filters:
    - Complaint Group
    - Neighborhood
    - Incident Zip
- Features:
    - Line chart (Altair) with hover tooltips
    - Multi-category comparison
    - Metrics:
        - Total complaints
        - Average monthly complaints
        - Peak month

Word Cloud Visualization

- Displays the most frequent 311 complaint types, where larger words indicate higher frequency within the selected filters
- Supports interactive filtering by neighborhood, ZIP code, complaint category, and specific issue, consistent with other pages
- Includes selectable color themes and a download option, along with summary metrics and a top complaints table for additional context

Key Findings

- Noise complaints are among the most frequent issues, especially in dense neighborhoods
- Sanitation and infrastructure issues show consistent presence across months
- Complaint patterns vary significantly by neighborhood and ZIP code
- Time trends reveal fluctuations, indicating possible seasonal or behavioral patterns
- Aggregating complaints into groups improves interpretability while maintaining detail

Conclusion

This project demonstrates how NYC 311 data can be transformed into an interactive analytical tool. By combining data cleaning, categorization, and visualization, we provide meaningful insights into urban issues affecting Manhattan residents.

The dashboard enables both high-level exploration and detailed analysis, making it useful for residents, researchers, and policymakers.